

widely used for industrial, social, corporate, military, and related applications. These are programmable libraries with support to customise as per the requirements of the domain in which to implement.

Robotic scenarios for industrial and corporate applications

CoppeliaSim is a cross-platform tool for simulation and industrial robotic applications. This tool makes possible to create the portable, scalable, and easy-to-maintain simulations for multiple scenarios. Its framework provides interfaces to various programming languages like C, Lua, Java, Python, MATLAB, and Octave.

For an interactive production environment, the CoppeliaSim Robot Simulator relies on a distributed control architectural system in which the individual control of each object or model requires an embedded script for high-performance control based applications. Such features give CoppeliaSim extremely versatile and adaptable powers for a broad range of robotic systems. CoppeliaSim is used



Fig. 1: Key dimensions of robotic deployments

in the development of high-speed algorithms, plant modeling, rapid prototyping, validations, remote control, double-check security, automated double-checking, and many others.

Following key features of CoppeliaSim make this platform highly effective and performance based for multiple applications including engineering, military, aerospace, health



Fig. 2: Official portal of Webots